TAFS API Documentation

Bill Inquiry & Bill Payment Integration
for United Bank Limited (UBL)

The American Foundation School (TAFS)
TAFS Development Team

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Purpose of This Document

This document describes the interface TAFS provides to United Bank Limited (UBL) for collecting school fees through UBL channels. It covers:

- the collection channels and how a payer identifies a student on each;
- the **Bill Inquiry** (`GET`) and **Bill Payment** (`POST`) APIs, with request and response samples for success and failure;
- the amount rules: within due date, after due date, and the 1Bill charges, which the payer bears;
- mandatory tracking of the **payment channel** and **payment method** on every payment;
- TAFS's stated consent to resolving every search by CC to the student's most recent voucher;
- the points on which TAFS asks UBL to confirm feasibility (Section 9).

Every payload in this document matches what the TAFS implementation sends and accepts.

1 Collection Channels

Channel	How the payer pays	Identifier entered
UBL branch (OTC)	Cash, cheque or pay order at any UBL branch counter.	The student's **CC number**.
UBL digital app	Bill Payment → select the **TAFS** beneficiary.	The student's **CC number**.
UBL internet banking	Bill Payment → select the **TAFS** beneficiary.	The student's **CC number**.
1Bill (any bank / wallet)	"1Bill Invoices" option in any participating bank or wallet.	The **1Bill consumer number** (Section 2.2).

These instructions are printed at the bottom of every TAFS fee voucher issued for collection into the UBL account.

2 Identifying a Student

2.1 CC number

Every TAFS student has a **CC number**, a numeric student code. To stay unambiguous as CCs reach 10,000, TAFS treats every CC as **5 digits, zero-padded**, and prints it that way on UBL vouchers (e.g. `08088` for CC 8088). The CC is the identifier for OTC, the UBL app

and internet banking. Leading zeros may be sent or omitted (`08088` and `8088` identify the same student).

2.2 1Bill consumer number

For 1Bill, the consumer number is a fixed **15-digit** number:

```
[10-digit TAFS 1Bill prefix][CC as 5 digits, zero-padded]
```

Examples, with prefix `1000000000`:

- CC `8088` → `100000000008088`
- CC `10001` → `100000000010001`

Request to UBL: 10-digit 1Bill prefix. TAFS requests UBL to issue (or arrange through 1Bill) the **10-digit 1Bill prefix** for The American Foundation School and share it with TAFS. Until it is received, TAFS vouchers and this document use the placeholder `1000000000`. Once UBL shares the prefix, TAFS will switch to it by configuration; no code change is needed on either side.

2.3 Voucher number

Vouchers collected through UBL carry a plain serial voucher number (e.g. `12461`), printed small at the top of the voucher. UBL does **not** need it to look up a bill. It is returned by the inquiry for reference, and UBL may echo it in the payment request (Section 7).

3 Voucher Resolution: Latest Voucher Rule & TAFS Consent

Every inquiry and payment made with a CC or a 1Bill consumer number resolves to **the most recent voucher generated by the school for that student**. Concretely, it is the newest voucher, excluding:

- vouchers the school has cancelled because a newer voucher replaced them; and
- vouchers the school has generated but not yet released to the parent.

If that voucher is already paid, has expired, or has been written off, the API says so and no older voucher is offered in its place.

Statement of consent. The American Foundation School confirms that it has requested this method and consents to it: every search by CC number (or by 1Bill consumer number), on every UBL channel, returns the most recent voucher generated by the school for that student as defined above. TAFS is satisfied that this is the correct bill to present and collect. TAFS accepts that UBL bears no responsibility for a payment being applied to that voucher rather than to any earlier voucher of the same student. Every voucher generated by the school carries the full amount the student owes, including any arrears.

4 Amount Rules

Situation	Amount payable
On or before the due date	The voucher amount (`amount_within_due_date`).
After the due date (until validity date)	Voucher amount + late payment surcharge, normally **PKR 1,000** (`amount_after_due_date`).
After the validity date	Not payable. The school must issue a new voucher (response code `04`).
Channel = `ONEBILL`	The amounts above **+ the 1Bill charges**, borne by the payer. This document assumes **PKR 30** until UBL confirms the actual charges (Section 9).
OTC, app, internet banking	The amounts above, with no addition.

- The inquiry returns `amount_payable`, the exact amount to collect *today* on the channel stated in the request.
- The payment must carry exactly that amount. Partial payments and overpayments are rejected (code `06`).
- TAFS wants the **parent to bear the 1Bill charges**: they are added on top of the fee amount shown to the payer. On a 1Bill payment TAFS records only the fee amount against the student; the 1Bill charge is noted against the payment and never credited to the student's account.
- TAFS can change the 1Bill charge by configuration as soon as UBL confirms it; no code change is needed.
- Dates are evaluated in Pakistan Standard Time (UTC+05:00).

5 Connectivity & Authentication

5.1 Base URL

```
https://tafs-backend-production.up.railway.app/api/v1/ubl
```

TAFS will also provide a separate test (UAT) base URL with test students before go-live.

5.2 Authentication

Every request must carry two HTTP headers. Credentials are issued by TAFS over a secure channel and are never sent in the URL.

Header	Type	Description
`X-UBL-User`	string	Service user ID issued to UBL
`X-UBL-Password`	string	Password for that service user
`Content-Type`	string	`application/json` (payment requests)

A request with missing or wrong credentials receives response code `10`. All traffic is HTTPS (TLS 1.2 or higher). TAFS can additionally restrict access to UBL's outbound IP addresses on request.

5.3 HTTP status

A well-formed, authenticated request always receives **HTTP 200**. The business outcome is in `response_code`. A malformed request receives **HTTP 400** with a description of the invalid field, for example a missing `channel` or a non-numeric `consumer_number`.

6 API 1 – Bill Inquiry (GET)

```
GET /api/v1/ubl/bill-inquiry?consumer_number={CC or 1Bill no}&channel={channel}
```

6.1 Query parameters

Parameter	Type	Required	Description
`consumer_number`	digits	Yes	CC number, up to 5 digits (OTC, app, internet banking), or 15-digit 1Bill consumer number.
`channel`	enum	**Yes**	`ONEBILL`, `OTC`, `APP` or `INTERNET_BANKING`. Decides whether the 1Bill charge is added.

6.2 Successful response fields

Field	Type	Description
`response_code`	string	`"00"`
`response_desc`	string	`"Success"`
`consumer_number`	string	As sent
`onebill_consumer_number`	string	The student's 15-digit 1Bill number
`cc`	string	Student CC number, 5 digits zero-padded
`student_name`	string	Student's full name
`gr_number`	string	School registration number
`campus`	string	Campus name
`class`	string	Class
`voucher_number`	string	Voucher being collected (plain serial)
`billing_month`	string	Month(s) billed, e.g. `"OCT 26"`
`voucher_status`	string	`UNPAID`, `OVERDUE` or `PARTIALLY_PAID`
`issue_date`	string	`YYYYMMDD`
`due_date`	string	`YYYYMMDD`
`validity_date`	string	`YYYYMMDD`; not payable after this date
`currency`	string	`"PKR"`
`amount_within_due_date`	number	Payable on or before the due date (incl. channel fee)
`amount_after_due_date`	number	Payable after the due date (incl. late fee and channel fee)
`late_fee`	number	Late payment surcharge applicable after the due date
`channel`	string	As sent
`channel_fee`	number	1Bill charge for `ONEBILL` (assumed `30`), else `0`
`amount_payable`	number	**Exact amount to collect today**

6.3 Sample – 1Bill inquiry before the due date

```
GET /api/v1/ubl/bill-inquiry?consumer_number=100000000008088&channel=ONEBILL
X-UBL-User: <issued by TAFS>
X-UBL-Password: <issued by TAFS>
```

```
{
  "response_code": "00",
  "response_desc": "Success",
  "consumer_number": "100000000008088",
  "onebill_consumer_number": "100000000008088",
  "cc": "08088",
  "student_name": "HAFSA DAHRI",
  "gr_number": "7001",
  "campus": "Gulistan-e-Jauhar Campus",
  "class": "AS",
  "voucher_number": "12461",
  "billing_month": "OCT 26",
  "voucher_status": "UNPAID",
  "issue_date": "20261001",
  "due_date": "20261010",
  "validity_date": "20261020",
  "currency": "PKR",
  "amount_within_due_date": 90030,
  "amount_after_due_date": 91030,
  "late_fee": 1000,
  "channel": "ONEBILL",
  "channel_fee": 30,
  "amount_payable": 90030
}
```


6.4 Sample – OTC inquiry after the due date

```
GET /api/v1/ubl/bill-inquiry?consumer_number=08088&channel=OTC
```


```
{
  "response_code": "00",
  "response_desc": "Success",
  "cc": "08088",
  "voucher_number": "12461",
  "voucher_status": "OVERDUE",
  "amount_within_due_date": 90000,
  "amount_after_due_date": 91000,
  "late_fee": 1000,
  "channel": "OTC",
  "channel_fee": 0,
  "amount_payable": 91000,
  ...
}
```


6.5 Sample – failure

A failed inquiry returns only `response_code`, `response_desc` and, where a voucher was found, `voucher_number`:


```
{ "response_code": "03", "response_desc": "Voucher is already paid", "voucher_number
    ": "12461" }
```


7 API 2 – Bill Payment (POST)

```
POST /api/v1/ubl/bill-payment
Content-Type: application/json
```

7.1 Request body

Field	Type	Required	Description
`consumer_number`	digits	Yes	Same value used in the inquiry
`channel`	enum	**Yes**	`ONEBILL`, `OTC`, `APP`, `INTERNET_BANKING`
`payment_method`	enum	**Yes**	How the payer paid. See Section 7.2
`amount`	number	Yes	Amount collected from the payer, PKR. Must equal `amount_payable` (incl. the 1Bill charge on 1Bill)
`transaction_id`	string	Yes	UBL's unique transaction reference (max 100). Used to prevent duplicate posting
`transaction_datetime`	ISO 8601	Yes	When the payment was taken, with offset, e.g. `2026-10-05T11:00:00+05:00`
`voucher_number`	string	Advised	`voucher_number` from the inquiry; guards against a newer voucher issued in between
`branch_code`	string	OTC	UBL branch code where cash/cheque was received
`payer_bank_code`	string	1Bill	Code of the bank or wallet the payer paid from
`payer_bank_name`	string	1Bill	Name of that bank or wallet
`instrument_number`	string	Cheque / PO	Cheque or pay-order number. Required when `payment_method` is `CHEQUE` or `PAY_ORDER`

7.2 Mandatory payment method tracking

TAFS requires every payment to state **both** the channel it arrived through and the method the payer actually used, so that each receipt can be reconciled. A payment without them is rejected (HTTP 400).

`payment_method`	Meaning
`CASH`	Cash at a branch counter
`CHEQUE`	Cheque (send `instrument_number`)
`PAY_ORDER`	Pay order / demand draft (send `instrument_number`)
`ACCOUNT_TRANSFER`	Debit from the payer's bank account (app, internet banking, 1Bill)
`DEBIT_CARD`	Debit card
`CREDIT_CARD`	Credit card
`WALLET`	Mobile wallet (e.g. via 1Bill)
`OTHER`	Any other method; please tell TAFS which so it can be added

7.3 Successful response

Field	Type	Description
response_code	string	"00"
response_desc	string	"Success"
transaction_id	string	As sent
tafs_receipt_id	string	TAFS receipt number for this payment
voucher_number	string	Voucher the payment was applied to
amount_booked	number	Amount credited to the student (excludes the 1Bill fee)
channel_fee	number	1Bill charge borne by the payer and retained by 1Bill (0 on other channels)

7.4 Sample – 1Bill payment

```
POST /api/v1/ubl/bill-payment
{
  "consumer_number": "100000000008088",
  "channel": "ONEBILL",
  "payment_method": "ACCOUNT_TRANSFER",
  "amount": 90030,
  "transaction_id": "UBL20261005110000123",
  "transaction_datetime": "2026-10-05T11:00:00+05:00",
  "voucher_number": "12461",
  "payer_bank_code": "HBL",
  "payer_bank_name": "Habib Bank Limited"
}
```

```
{
  "response_code": "00",
  "response_desc": "Success",
  "transaction_id": "UBL20261005110000123",
  "tafs_receipt_id": "48215",
  "voucher_number": "12461",
  "amount_booked": 90000,
  "channel_fee": 30
}
```

7.5 Sample – OTC cheque payment after the due date

```
{
  "consumer_number": "08088",
  "channel": "OTC",
  "payment_method": "CHEQUE",
  "instrument_number": "00412387",
  "branch_code": "0457",
  "amount": 91000,
  "transaction_id": "UBLOTC0457202610121015",
  "transaction_datetime": "2026-10-12T10:15:00+05:00",
  "voucher_number": "12461"
}
```

7.6 Sample – failures

```
{ "response_code": "06", "response_desc": "Amount does not match the payable amount
    ", "amount_payable": 90030 }
{ "response_code": "08", "response_desc": "Duplicate transaction; already posted",
  "transaction_id": "UBL20261005110000123", "tafs_receipt_id": "48215" }
```

7.7 Duplicates, retries and timeouts

- Posting is idempotent on `transaction_id`. If UBL does not receive a response (timeout), it should resend the **same** request with the **same** `transaction_id`. A payment that was already posted returns code `08` with the original `tafs_receipt_id` and is never booked twice.
- Code `08` therefore means the payment **is** recorded at TAFS and should be treated as a success.
- Code `99` means nothing was booked; UBL may retry.

8 Response Codes

Code	Meaning	Used by
`00`	Success	Inquiry, Payment
`01`	Consumer number not found	Inquiry, Payment
`02`	No voucher has been issued for this student	Inquiry, Payment
`03`	Voucher is already paid	Inquiry, Payment
`04`	Voucher validity has expired; school must issue a new voucher	Inquiry, Payment
`05`	Voucher is not payable (written off)	Inquiry, Payment
`06`	Amount does not match the payable amount	Payment
`07`	A newer voucher has been issued; inquire again	Payment
`08`	Duplicate transaction; already posted (treat as success)	Payment
`09`	Invalid request (e.g. cheque without `instrument_number`)	Payment
`10`	Invalid credentials	Inquiry, Payment
`99`	General error; nothing booked, retry later	Inquiry, Payment

9 Points for UBL to Confirm

TAFS asks UBL to confirm the following before integration testing:

1. **1Bill channel on inquiry.** Can UBL tell TAFS, on the `GET` inquiry itself, that the request came through 1Bill (the `channel=ONEBILL` parameter), so that the 1Bill charge is included in the amount shown to the payer? If 1Bill inquiries reach TAFS in a different way, for example through a separate 1Bill-facing endpoint or with different credentials, please describe it and TAFS will adapt.

2. **1Bill charges (please specify).** What are the 1Bill charges on a TAFS bill payment? Please state whether it is a flat amount or a percentage, any slabs or caps, and whether it differs by paying bank or wallet. TAFS wants the **parent to bear these charges**, added on top of the fee. Please confirm that 1Bill collects them from the payer and retains them, and that only the fee amount is settled into the TAFS account. The PKR 30 used in this document is an assumption pending your answer.

3. **Payment method on the payment request.** Can UBL send `payment_method` on every payment, for every channel, using the values in Section 7.2? For 1Bill, can UBL also pass the paying bank or wallet (`payer_bank_code` / `payer_bank_name`)?

4. **OTC search by CC.** Can the OTC teller screen search by CC number and display the inquiry result before accepting payment?

5. **Cheques.** Are cheques posted to TAFS when deposited, or only once cleared? If posted at deposit, TAFS will need a way to be told about a returned cheque (see item 7).

6. **1Bill prefix (requested).** Please issue TAFS's 10-digit 1Bill prefix (Section 2.2) and confirm when it will be live.

7. **Reversals.** This version has no reversal API. A payment posted in error is corrected by the school on request. Please tell TAFS if UBL requires an online reversal message, and its format.

8. **Reconciliation.** Can UBL share a daily settlement statement listing `transaction_id`, amount, channel and payment method, so TAFS can reconcile against its receipts?

9. **Message format.** If UBL requires its own standard message format (field names, fixed-length amounts, response codes), please share the specification and TAFS will map to it.

10 What Is Printed on the Voucher

Each voucher issued for collection into the UBL account carries, at the bottom:

- **UBL branches:** pay at any UBL branch by quoting the student's CC number.
- **UBL digital app:** go to Bill Payment, select "TAFS" as the beneficiary and enter the student's CC number.
- **Other banks / digital channels (1Bill):** pay via "1Bill Invoices" using the 1Bill consumer number printed on the voucher; 1Bill service charges are added to the bill and paid by the payer.

For The American Foundation School	For United Bank Limited
Name / Designation / Date	Name / Designation / Date